

Sophia N. Wilson

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Biography

Sophia N. Wilson is a Ph.D. fellow at the Sustainable Artificial Intelligence for Sciences (SAINTS) Lab, Department of Computer Science, University of Copenhagen (UCPH), supervised by Raghavendra Selvan and Erik Dam. Her research spans two strands: a technical strand developing algorithms and AI systems that are more resource efficient in data, compute, and energy use; and a socio-technical strand measuring and mitigating sustainability harms in the AI industry, including environmental impact and poor labour conditions. She is also active in bridging academia and industry, and in outreach promoting STEM fields, particularly to girls and young women.

Education

- 2025–present **PhD Fellow, Sustainable Artificial Intelligence, SAINTS Lab, Dept. of Computer Science, UCPH, Denmark.**
Resource-efficient machine learning (ML), sustainable AI.
- 2023–2025 **Master in Computational Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 12/12. Physics-informed ML, applied ML, applied statistics, scientific programming, cosmology.
- 2019–2023 **Bachelor in Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 11/12. Computational physics, statistics, numerical methods, astrophysics, cosmology.

Publications

In Conference Proceedings

- 2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, 2026.
- 2026 **Sophia N. Wilson**, Guðrún Fjóla Guðmundsdóttir, Andrew Millard, Raghavendra Selvan, and Sebastian Mair. Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, 2026.
- 2026 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Performance-Carbon Trade-Offs across Architectural Biases in Shear Flow Forecasting. In *Proceedings of the 19th European Conference on Computer Vision (ECCV), LIMIT Workshop*, 2026.

Preprints

- 2026 Pedram Bakhtiarifard, **Sophia N. Wilson**, Mahmoud Afifi, Jonathan Wenshøj, and Raghavendra Selvan. Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis, 2026.

Research Projects

Dept. of Computer Science, UCPH

- Spring 2026 **Connecting Algorithmic Complexity and Deep Neural Networks.**
Collaborative work developing QuBD (Quantized Block Decomposition), a scalable method for estimating the algorithmic complexity of deep neural network weights, providing the first scalable characterisation of the learning-as-compression hypothesis for deep neural networks.
- Fall 2025 **Costs of Data Abundance and Benefits of Data Frugality in AI.**
Led two studies: (1) quantified the environmental, social, and economic costs of hyper-datafication through large-scale metadata analysis of 550,000 Hugging Face datasets, a survey of data worker conditions in Kenya, and an examination of data centre investment and geographic concentration, accepted at FAccT 2026; (2) identified a preach-practice gap in the data frugal literature where energy savings are frequently cited as motivation but rarely measured, and developed recommendations for closing it, accepted as a position paper at ICML 2026.
- Niels Bohr Institute & Dept. of Computer Science, UCPH
- Fall, 2024 – **Quantifying the Reduction in Carbon Footprint using Physics-Informed ML.**
Spring, 2025 Completed my MSc thesis applying physics-informed ML to reduce energy consumption in PDE simulations.

International Experience

- Aug. 2026 Eastern European Machine Learning Summer School, **Cetinje, Montenegro.** (Best Poster).
- Feb. 2026 Machine Learning Summer School, **Melbourne, Australia.**
- Spring 2024 Exchange semester at the **Institute of Physics, University of Amsterdam, The Netherlands.**
- Aug. 2022 Astromatic Machine Learning Summer School at the **University of Montréal, Canada.**
- Aug. 2021 Nordic Optical Telescope Summer School, **La Palma, Spain.**

Teaching Experience

- 2026 **Data Analysis and Machine Learning (for high school teachers),** Dept. of Computer Science, Aarhus University.
- 2025 **Machine Learning A,** Dept. of Computer Science, UCPH.
- 2025 **Mathematics A,** Falkonergårdens Gymnasium.
- 2023 **Summer Science Camp,** Insero.
- 2022-23 **Mathematics for Physicists 2,** Niels Bohr Institute, UCPH.
- 2022-23 **Mathematics for Physicists 1,** Niels Bohr Institute, UCPH.
- 2020-21 **Mechanics and Special Relativity,** Niels Bohr Institute, UCPH.

Supervision

- 2026 **Co-supervising two BSc students,** Faculty of Humanities, UCPH.

Invited Talks and Panel Debates

- May 2026 **Speaker at P1 Workshop on Green AI,** Pioneer Center for AI, UCPH.
- Apr. 2026 **Speaker and panellist at Sustainability Week,** Lund Stem Cell Center, Lund University.
- Sep. 2025 **Speaker at Resource-Efficient AI Workshop,** D3A Conference.
- Apr. 2025 **Speaker at Climate Action Day,** Copenhagen Center for Social Data Science, UCPH.

Guest Lecturing, Workshops, and Outreach

- 2026 **Presenting Hyper-Datafication paper,** Visual Computing, Technical University of Denmark.
- 2026 **Presentation at AI and Sustainability Workshop by the Tech Policy Youth Committee,** The National Center for AI in Society, UCPH.

- 2026 **Presenting Data Frugality paper**, *Embedded AI Group*, Technical University of Denmark.
- 2026 **Co-organiser and moderator at the SAINTS'26 workshop on Sustainable AI**, *SAINTS Lab, Dept. of Computer Science*, UCPH.
- 2026 **Guest-lecturing on AI, Resource Use and Sustainability**, Københavns Professionshøjskole.
- 2026 **Organiser of a workshop on Environmental Impacts of AI**, Det Åbne Gymnasium.
- 2025 **Guest-lecturing in PhD course on Machine Learning for Sciences**, *Dept. of Computer Science*, UCPH.
- 2025 **Talks about Studying Physics at UCPH**, Falkonergårdens Gymnasium.
- 2025 **Presentation of current MSc thesis work at the 4th Annual Niels Bohr Institute MSc Student Symposium**, *Niels Bohr Institute*, UCPH.
- 2021-24 **Member of Women in Physics contributing to initiatives promoting gender balance in STEM**, *Niels Bohr Institute*, UCPH.

Referees

Raghavendra Selvan

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Dept. of Computer Science
UCPH

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Jens Hesselbjerg Christensen

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